[scan 331]

$$
\begin{array}{ccccccccc}
1 & \to & \pi_1(M_{g,\nu\overline{\mathbb{Q}}}) & \longrightarrow & \pi_1(M_{g,\nu}) & \longrightarrow & \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) & \longrightarrow & 1 \\
 & & \downarrow{\scriptstyle |^2} & & \downarrow & & \downarrow & & \\
1 & \longrightarrow & \Sigma & \longrightarrow & \mathcal{N}|N_\Sigma & \longrightarrow & \Gamma_\Sigma & \longrightarrow & 1 \\
 & & & & \| & & & \searrow & \\
 & & & & \mathrm{Norm}_{\widehat{\mathcal{F}}(\pi)}(\Sigma) & & & & \\
\end{array}
\tag{26}
$$

m1

*Marginal note.* m1 group of "arithmeticouter "arithmetic" automorphisms of $\pi$ (endowed with $\Sigma$)

[scan 331]

$$
\begin{array}{ccccccccc}
1 & \to & \pi_1(M_{g,\nu\overline{\mathbb{Q}}}) & \longrightarrow & \pi_1(M_{g,\nu}) & \longrightarrow & \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) & \longrightarrow & 1 \\
 & & \downarrow{\scriptstyle \jmath|2} & & \downarrow & & \downarrow & & \\
1 & \longrightarrow & \Sigma & \longrightarrow & \mathcal{N}|N_\Sigma & \longrightarrow & \Gamma_\Sigma & \longrightarrow & 1 \\
 & & & & \| & & & \searrow & \\
 & & & & \mathrm{Norm}_{\widehat{\mathcal{F}}(\pi)}(\Sigma) & & & & \\
 & & & & & & & & {\scriptstyle \mathrm{m2}}
\end{array}
\tag{26}
$$

*Marginal note.* [m2] group of [d1]outer[i1] "arithmetic" automorphisms of $\pi$ (endowed with $\Sigma$)

$$
\begin{array}{ccccccccc}
1 & \to & \pi_1(M_{g,\nu\overline{\mathbb{Q}}}) & \longrightarrow & \pi_1(M_{g,\nu}) & \longrightarrow & \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) & \longrightarrow & 1 \\
 & & \downarrow{\scriptstyle |\cdot|^2} & & \downarrow & & \downarrow & & \\
1 & \longrightarrow & \Sigma & \longrightarrow & \mathcal{N}|N_\Sigma & \longrightarrow & \Gamma_\Sigma & \longrightarrow & 1 \\
 & & & & \| & & & \searrow & \\
 & & & & \mathrm{Norm}_{\widehat{\mathcal{F}}(\pi)}(\Sigma) & & & & \\
 & & & & & & & & \text{m3}
\end{array}
\tag{26}
$$

*Marginal note.* [m3] group of †outer "arithmetic" automorphisms of $\pi$ (endowed with $\Sigma$)